Characterizing Extended Kalman Filter Fusion of Visual Detection and Odometry Under Sparse Observation: A Multi-Robot Digital Twin Case Study

Anonymous Author(s)

Institution withheld for double-blind review

Abstract. Multi-robot systems combining odometry with external visual detection often rely on a discrete fallback — trusting detection when available, odometry otherwise — which discards uncertainty information and offers no principled behavior during long detection gaps. We replace this fallback with a per-robot Extended Kalman Filter fusing odometry and YOLO-based visual detection from a fixed overhead camera, validated in two stages (synthetic NEES/NIS consistency, then real recorded data) on a ROS2/Gazebo digital twin using three TurtleBot3 robots. Measured at correction instants, the filter reduces position error by +23.7% relative to odometry alone under aggressive drift. However, measured continuously — at every odometry reading — the same filter is 12.7% *worse* than raw odometry, under the sparse visual coverage (0–28%) our camera setup exhibits in practice. We show this reversal is the expected consequence of operating below a critical observation rate, not a design flaw, and connect it mechanistically to a covariance ceiling required to keep Mahalanobis-gated association stable. We argue that reporting both metrics, rather than only the commonly reported correction-instant gain, is necessary to honestly characterize EKF fusion under sparse observation, and discuss why alternative filters (UKF, particle filter, IMM) do not resolve the underlying limitation.

Keywords: Extended Kalman Filter · Sensor Fusion · Multi-Robot Localization · Sparse Observations · Digital Twin

1 Introduction

Multi-robot systems combining onboard odometry with external visual detection face a recurring choice: how to combine two noisy position estimates arriving at different rates, not always available at the same instant. A common, pragmatic solution — used in the digital twin studied here prior to our contribution — is a *discrete fallback*: trust detection when confidently associated, odometry otherwise. This is simple and robust but discards information: no partial correction from low-confidence detections, no uncertainty propagation, and no principled behavior during long detection-free stretches.

Extended Kalman Filtering (EKF) is the classical answer, offering full probabilistic prediction/correction with explicit uncertainty propagation. Its convergence properties, however, are classically established under reasonably dense, regular observation [2] — an assumption a single fixed overhead camera, subject to finite field of view and self-occlusion during maneuvers, does not generally satisfy. Whether this matters in practice, on a real platform rather than only in simulation, is the question this paper investigates.

We replace the discrete fallback with a per-robot EKF fusing odometry and YOLO-based visual detection, evaluated end-to-end on a ROS2/Gazebo digital twin with three TurtleBot3 robots. Our contribution is twofold. First, validated in two stages (synthetic NEES/NIS, then real recorded data), the filter delivers a real fusion gain (+23.7% position error reduction relative to odometry alone, under aggressive drift, at correction instants). Second, and more centrally, the same filter measured continuously — not only at correction instants — is 12.7% *worse* than raw odometry under the sparse visual coverage (0–28%) our camera exhibits in practice. We explain this reversal mechanistically, connecting it to established theory on Kalman filtering under intermittent observations [3,4], and argue that reporting both metrics — not only the commonly reported correction-instant gain — is necessary for honest characterization, and discuss why alternative filters do not resolve the underlying limitation.

2 Related Work

Bailey and Durrant-Whyte [1] formalize the standard EKF motion/observation model our filter follows directly (odometry drives prediction, YOLO detections drive correction; notation adopted in Section 3). Dissanayake et al. [2] prove EKF-SLAM covariance convergence under *dense* landmark observation — an assumption our 0–28%-coverage overhead camera violates by construction, which is central to Section 6. Sinopoli et al. [3] establish that, under Bernoulli observation arrival, covariance diverges below a critical rate; our Scenario 3 (0% coverage) sits at the limiting case $p \to 0$, so the saturation we observe is theoretically expected, not a symptom of misconfiguration. Censi [4] notes this critical threshold depends on the covariance summary metric used; we report trace(P) as a scoping choice, not a claim of metric-independence.

Khosravi et al. [7] report decentralized cooperative localization for two robots (mutual LiDAR corrections, Gazebo and hardware), with 34–56% RMSE reduction — comparable in order of magnitude to our +23.7%. Ahmed and Frémont [8] fuse a custom YOLOv8-RGBDis detector with a per-target EKF for a maritime multi-robot team, reporting mAP@0.5 = 0.995, numerically close to ours despite different architecture/domain. Neither reports a continuous-error regime in which fusion underperforms the unfused baseline — the differentiating finding of this work. Dobrynin and Garcia [9] share our exact platform (three TurtleBot3, ROS2) but solve camera-free relative localization, so do not encounter sparse visual coverage; we revisit this as complementary work in Section 7. Our contribution is the joint report of both the correction-instant gain and the continuous-error degradation, grounded in [3,4] for why this trade-off is expected rather than an implementation bug.

3 Methodology

Following Bailey and Durrant-Whyte [1], each robot is tracked by an independent EKF with state $x_k = [p_x, p_y, \theta]^\top$. The motion model integrates the *delta* of consecutive odometry readings onto the current state — not the absolute value, which would discard prior visual corrections. The observation model maps state to the 2D position reported by a YOLO detector on a fixed overhead camera (Section 4).

Prediction and covariance capping. Process noise is $Q = \mathrm{diag}(0.5, 0.5, 0.25)$, scaled by $\Delta t/0.1$ for the real odometry rate ($\approx$29 Hz vs. the 0.1s assumed during synthetic calibration); calibrated empirically against real data (a sweep from 10^{-6} to 1.0 showed monotonically increasing gain up to saturation at $Q \geq 0.5$). A hard ceiling $\mathrm{clip}(P, -\infty, 0.05)$ (element-wise) is applied after each prediction: without it, P grew unbounded during long uncorrected intervals (observed $\mathrm{trace}(P) \approx 85$ vs. an expected ≈ 0.5), breaking Mahalanobis-gated association by widening a robot's acceptance region enough to claim a neighbor's detection. This is functionally analogous to *track coasting* in multi-target tracking (Bar-Shalom), and theoretically motivated by Sinopoli et al.'s [3] covariance-divergence result — it trades strict optimality for bounded, predictable behavior under the sparse-observation regime we operate in. We report the ceiling's effect on continuous-time error explicitly in Section 5, not as an implementation detail.

Correction. Measurement noise R is derived linearly from YOLO detection confidence $c \in [0, 1]$ (carried in the otherwise-unused z component of each detected pose), $\sigma = R_{\min} + (1 - c)(R_{\max} - R_{\min})$, $R_{\min} = 0.02$, $R_{\max} = 0.20\,\mathrm{m}$ — a high-confidence detection is trusted more during the Kalman update.

Data association. With up to three robots visible simultaneously, each detection is assigned via Mahalanobis-gated nearest-neighbor: $d^2 = e^\top P_{xy}^{-1} e$ per (robot, detection) pair, accepted only if $d^2 \leq 9.21$ (99% confidence, χ^2, 2 DOF). This inherits a known limitation (Altendorfer and Wirkert [6]): an inflated covariance widens the acceptance region and can bias association toward more uncertain tracks — the same failure mode motivating the covariance ceiling above.

Camera calibration. Intrinsics come from `cv2.calibrateCamera` (Zhang [5]) using a synthetic checkerboard at multiple poses, with a sanity check rejecting low-reprojection-error but ill-conditioned solutions (focal length far from the FOV-derived theoretical value) — accepted calibration: $f_x = 739.4$, $f_y = 733.6$, mean reprojection error 0.162px.

Two-stage validation. (1) Synthetic NEES/NIS: 30 Monte Carlo seeds (2000 steps each) give NEES$= 2.955$ (expected 3.0) and NIS$= 1.982$ (expected 2.0), within 1.5%/0.9% relative error — well-calibrated in practice despite formal χ^2 95% bounds being too narrow to pass at this sample size. (2) The calibrated filter is then run against real odometry and YOLO detections recorded from the live digital twin, where observation coverage is sparse and irregular by construction of the physical scenario, not by design choice.

4 Experimental Setup

We evaluate our EKF in a ROS2/Gazebo digital twin: three TurtleBot3 Waffle units in a shared simulated world alongside a static overhead camera at $(0, 0, 3)$ m, 90° downward pitch, 60° horizontal FOV, publishing 640×480 frames at 10 Hz. Each robot also publishes odometry (≈29 Hz) and an unused 2D LiDAR scan (Section 7). Robot detection uses a YOLO model trained on 375 annotated frames from the same camera, mAP@0.5 = 0.995 on held-out validation; confidence is carried in the detection message's otherwise-unused z field for direct consumption by the EKF correction step.

Three ∼18 s scenarios were recorded as ROS2/MCAP bags: (a) **Stationary** — 152 detections vs. 550 odometry readings/robot, ≈28% coverage, the highest of the three; (b) **Straight motion** ($v_x = 0.1$ m/s) — 144 vs. 522 readings, ≈11% coverage, ∼1.5 m confirmed ground-truth displacement; (c) **Curve/occlusion** ($v_x = 0.12$ m/s, $\omega_z = 0.4$ rad/s) — 145 detections recorded on the topic, but **zero** successfully associated to `robot1` over the full recording: the single fixed viewpoint loses robots to self-occlusion and out-of-frame motion during turns, producing the 0% coverage case central to Section 5. Timestamp synchronization between detections and odometry was verified via `rosbag2_py` in all three scenarios (no gap >500 ms on any channel), ruling out sensor-clock desynchronization as a confound.

We release a reproducibility package (world file, URDF, calibration, EKF hyperparameters, code commit SHA) supporting independent verification, and document one non-obvious dependency: `ros-humble-rosbag2-storage-mcap` is not installed by default on ROS2 Humble.

5 Results

We report two distinct error metrics, deliberately kept separate rather than collapsed into a single number: *correction-instant error*, measured only at the moments a YOLO detection is successfully associated and fused (the metric most commonly reported in related work — Section 2), and *continuous error*, measured at every odometry reading regardless of whether a correction occurred. As we show below, these two metrics disagree in sign, and both are necessary to honestly characterize the filter's behavior under sparse observation.

5.1 Filter Calibration

The synthetic NEES/NIS validation described in Section 3 confirms the filter is well-calibrated before any real-data evaluation. Fig. 1 shows both statistics remaining close to their theoretical expected values (NEES→ 3.0, NIS→ 2.0) across the Monte Carlo run.

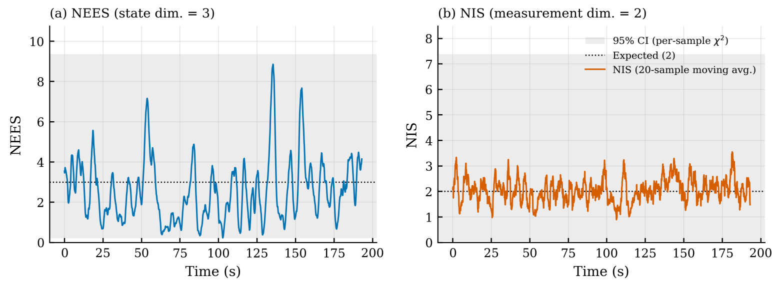

Fig. 1. NEES and NIS consistency over the synthetic Monte Carlo validation (30 seeds, 2000 steps each). Both statistics track their theoretical expected values, indicating a well-calibrated filter.

5.2 Correction-Instant Error

Table 1 reports position error under three injected odometry drift regimes (no drift, light drift, aggressive drift), aggregated across all three recorded scenarios ($n = 1323$). Under aggressive drift — the regime in which raw odometry has already accumulated substantial error before a visual correction becomes available — the EKF reduces mean position error by +23.7% relative to odometry alone (Wilcoxon signed-rank, $p < 0.001$). Under light drift, the gain is negative (-5.3%): visual detection error (≈3–6 cm) is not yet smaller than the still-small odometry error, so fusion offers no benefit in this regime — expected behavior for a correctly calibrated filter, not a failure.

Table 1. Correction-instant position error (m), aggregated over 3 scenarios ($n = 1323$).

Drift regime	**EKF (mean)**	**Odom-only (mean)**	**Change**
None (odom = GT by construction)	0.0356	0.0000	N/A
Light	0.0528	0.0501	-5.3%
Aggressive	0.1282	0.1681	**+23.7%**

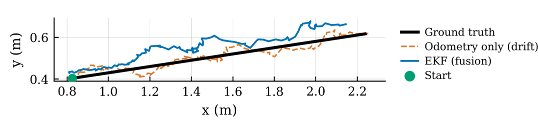

Fig. 2. Estimated trajectory under aggressive drift, Scenario 2 (straight motion). The EKF (blue) tracks ground truth (black) more closely than raw drifted odometry (orange, dashed).

Fig. 2 shows this qualitatively: raw odometry visibly drifts from the straight ground-truth path, while the EKF estimate tracks it more closely.

5.3 Continuous Error: The Central Finding

When error is instead measured at *every* odometry reading — not only at correction instants — the result reverses. Table 2 reports continuous-time error under aggressive drift, using the same recordings as Table 1. Aggregated across all readings ($n = 1607$), the EKF is 12.7% *worse* than raw odometry (Wilcoxon signed-rank, $p < 0.001$) — the opposite sign from the correction-instant result.

Table 2. Continuous position error (m), every odometry reading, aggressive drift.

Scenario	YOLO coverage	EKF (mean)	Odom-only (mean)	Change
Stationary	27.6%	0.2116	0.1774	−19.3%
Straight motion	11.3%	0.2115	0.1780	−18.8%
Curve / occlusion	0.0%	0.1778	0.1778	−0.0%
Aggregated	—	**0.2003**	**0.1777**	**−12.7%**

Two observations explain this reversal. First, the degradation scales inversely with YOLO coverage: Scenario 3 (0% coverage — no successful visual correction over the entire ~18 s recording) shows a change of exactly 0.0%, because with zero corrections the EKF's prediction step degenerates to pure odometry-delta integration, making it mathematically identical to the unfused baseline. Scenarios 1 and 2 (27.6% and 11.3% coverage) show similar ~19% degradation despite their different coverage levels, suggesting the effect saturates quickly once corrections become sparse enough for the covariance ceiling (Section 3) to bind.

Second, and mechanistically explaining the first, Fig. 3 shows why: trace(P) saturates at the covariance ceiling almost immediately after any gap without correction, and only drops when a correction lands. In Scenario 3, the trace saturates within the first observed step and never drops again for the remainder of the recording. This is precisely the divergence regime predicted by Sinopoli et al. [3] for observation arrival rates below a critical threshold: the filter's covariance ceiling prevents unbounded growth (and the association failures it would otherwise cause, Section 3), but it does so by capping the filter's *own estimate of its uncertainty*, not by making its state estimate actually more accurate during the uncorrected interval — the filter remains overconfident in its odometry-only prediction relative to how uncertain that prediction actually is.

6 Discussion

The natural reading of Table 2 is unflattering: our EKF, evaluated continuously, loses to the odometry it should improve upon. We argue this is expected, not a

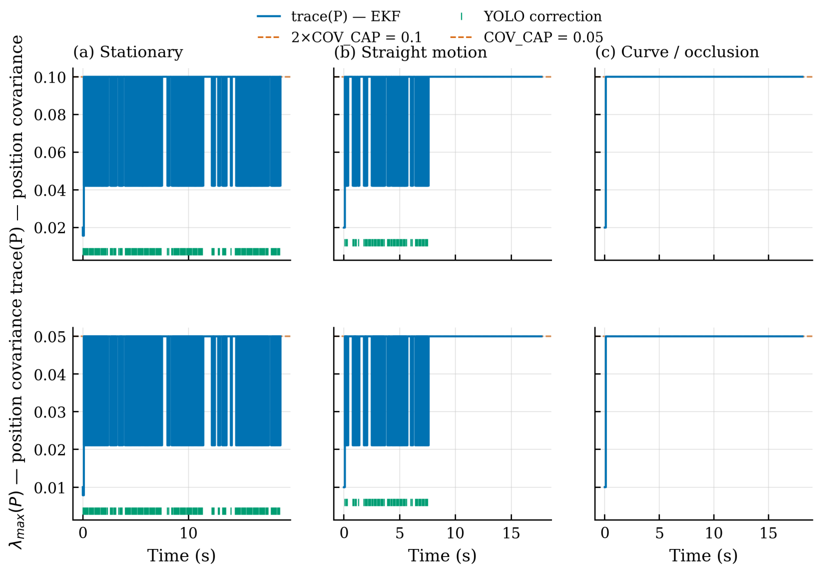

Fig. 3. Position-covariance saturation over time (step plots), all three scenarios, under two metrics (top: trace(P_{xy}); bottom: $\lambda_{max}(P_{xy})$, the largest eigenvalue), following Censi's [4] observation that the critical observation threshold depends on the chosen error metric. Short vertical ticks along the bottom of each panel mark successful YOLO corrections. Dashed lines mark each metric's ceiling under the same element-wise clip (Section 3): $2 \times \text{COV\_CAP} = 0.1$ for the trace (sum of both eigenvalues under near-isotropic P_{xy}) and $\text{COV\_CAP} = 0.05$ for the largest eigenvalue alone. Both metrics agree qualitatively: coverage decreases left to right ($27.6\% \rightarrow 11.3\% \rightarrow 0\%$), and covariance saturates at its ceiling for correspondingly longer stretches, reaching permanent saturation in the zero-coverage case.

flawed design. Dissanayake et al.'s [2] convergence proof assumes dense observation; our camera violates that by construction (0–28% coverage), and Sinopoli et al. [3] predict exactly this covariance divergence without a ceiling, and resulting overconfidence with one. Scenario 3's 0.0% change is a clean confirmation: with zero corrections, the filter's prediction step is provably identical to odometry-delta integration. We follow Censi [4] in reporting trace(P) as a scoping choice, not a claim of metric-independent divergence.

No standard alternative resolves the zero-coverage case: a particle filter degenerates to pure process propagation with no likelihood to reweight against; a UKF targets observation-model nonlinearity, not observation *rate* (our near-linear projection gains little); an IMM filter still requires some observation to weight competing modes. We treat low coverage — a single fixed viewpoint subject to occlusion — as the root cause, consistent with sensor-placement literature: estimation limits under a given sensor topology hold independently of the filter used. The covariance ceiling is thus an engineering approximation of *track coasting* (Bar-Shalom), trading optimality for association stability, and inherits a known gating limitation (Altendorfer and Wirkert [6]): a saturated covariance widens the acceptance region precisely when true uncertainty is least well characterized by a fixed value — though we did not observe incorrect cross-robot association in our three scenarios.

7 Future Work

Concrete directions include: (a) **adaptive process noise** — scale Q by time since the last correction instead of a fixed ceiling, more principled at the cost of an added hyperparameter; (b) **LiDAR as a third fusion source** — each robot already publishes unused 2D scans; mutual LiDAR corrections (in the spirit of Khosravi et al. [7]) could target the zero-coverage regime directly; (c) **formal projection uncertainty** — propagate the full projection Jacobian (calibration error, ground-plane assumption) into R; (d) **combining with distributed relative localization** — Dobrynin and Garcia's [9] camera-free distributed filtering on the same platform could provide a second correction channel available during visual-coverage gaps.

8 Conclusion

We replaced a discrete visual/odometry fallback with a two-stage validated Extended Kalman Filter, evaluated on real recorded data from three TurtleBot3 robots. The filter delivers a genuine gain at correction instants (+23.7%) but is measurably worse than raw odometry when evaluated continuously (−12.7%) under 0–28% visual coverage — including a zero-coverage scenario collapsing, exactly as expected, to the unfused baseline. Rather than report only the favorable metric, we characterize both and ground the reversal in established theory on intermittent-observation Kalman filtering. EKF fusion remains the right tool for this problem, but its correctness under sparse observation must be measured and reported explicitly, and the observation rate itself — not the choice of filter — is the binding constraint future work should target.

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